Task Submission: Console-Based Mini Project

1. Execution Summary

The objective of this task was to develop a functional, console-based application to demonstrate proficiency in Java fundamentals, specifically handling user input, control structures, and array management. The "Academic Record Management System" was designed to serve as a streamlined tool for educational data entry and retrieval.

The system provides a robust interface for administrators to register students, calculate academic standing based on examination marks, and search for specific records within the database.

2. System Module: Academic Record Management

This module serves as the core of the application, utilizing a menu-driven interface to perform the following operations:

- **Data Registration:** The system captures official student names and marks via the Scanner class, storing them in parallel arrays for persistent access within the session.
- **Grading Logic:** A multi-level conditional structure evaluates raw scores and assigns formal grades (A+, A, B, C, F) rather than a simple pass/fail metric, providing more granular data.
- **Search Functionality:** A search algorithm was implemented using a for loop and String comparison methods. This allow for the immediate retrieval of specific student data from the array based on user queries.
- **Capacity Management:** The system includes error-handling logic to prevent data overflow by monitoring the MAX_STUDENTS constraint.

3. Conclusion

The **Academic Record Management System** successfully integrates the primary building blocks of Java development: user input processing, iterative loops, and conditional decision-making. By utilizing a modular approach, the code remains readable and easily maintainable. All system functionalities, including registration, grade calculation, and data searching, have been verified through rigorous testing and successful execution.

Source Code (Student.java)

```java
import java.util.Scanner;


public class Stundent {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);


        // System Constraints
        final int MAX_STUDENTS = 10;
        String[] studentNames = new String[MAX_STUDENTS];
        int[] studentMarks = new int[MAX_STUDENTS];
        int currentCount = 0;


        System.out.println("====================================
==========");
        System.out.println("     ACADEMIC RECORD MANAGEMENT
SYSTEM        ");
```

```
        System.out.println("==============================================");

        while (true) {
            System.out.println("\nMAIN MENU:");
            System.out.println("1. Register New Student");
            System.out.println("2. View All Academic Records");
            System.out.println("3. Search Student by Name");
            System.out.println("4. Terminate System");
            System.out.print("Please select an option: ");

            int action = scanner.nextInt();
            scanner.nextLine(); // Clear buffer

            if (action == 1) {
                if (currentCount < MAX_STUDENTS) {
                    System.out.print("Enter Official Name: ");
                    studentNames[currentCount] = scanner.nextLine();

                    System.out.print("Enter Examination Marks (0-100): ");
                    studentMarks[currentCount] = scanner.nextInt();

                    currentCount++;
                    System.out.println("SUCCESS: Record synchronized.");
```

```
            } else {
                System.out.println("ERROR: Database capacity reached.");
            }
        }
        else if (action == 2) {
            if (currentCount == 0) {
                System.out.println("NOTICE: No data currently 
available.");
            } else {
                System.out.println("\n--- COMPREHENSIVE STUDENT 
RECORDS ---");
                for (int i = 0; i < currentCount; i++) {
                    String grade;
                    // Formal grading logic
                    if (studentMarks[i] >= 90) grade = "A+";
                    else if (studentMarks[i] >= 75) grade = "A";
                    else if (studentMarks[i] >= 50) grade = "B";
                    else if (studentMarks[i] >= 35) grade = "C (Pass)";
                    else grade = "F (Fail)";

                    System.out.println("ID: " + (i + 1) + " | Name: " + 
studentNames[i] +
                                        " | Marks: " + studentMarks[i] + " | Grade: " 
+ grade);
                }
```

```
            }
        }
        else if (action == 3) {
            System.out.print("Enter name to search: ");
            String query = scanner.nextLine();
            boolean found = false;

            for (int i = 0; i < currentCount; i++) {
                if (studentNames[i].equalsIgnoreCase(query)) {
                    System.out.println("MATCH FOUND: " +
studentNames[i] + " | Marks: " + studentMarks[i]);
                    found = true;
                    break;
                }
            }
            if (!found) System.out.println("ERROR: Record not found.");
        }
        else if (action == 4) {
            System.out.println("System shutting down. Secure logout
complete.");
            break;
        }
        else {
            System.out.println("WARNING: Invalid selection. Please try
again.");
```

```
            }
        }
        scanner.close();
    }
}
```

Output:

```
MAIN MENU:
1. Register New Student
2. View All Academic Records
3. Search Student by Name
4. Terminate System
Please select an option: 2

--- COMPREHENSIVE STUDENT RECORDS ---
ID: 1 | Name: Student1 | Marks: 90 | Grade: A+
ID: 2 | Name: Student2 | Marks: 80 | Grade: A
ID: 3 | Name: Student3 | Marks: 70 | Grade: B

MAIN MENU:
1. Register New Student
2. View All Academic Records
3. Search Student by Name
4. Terminate System
Please select an option: 4
System shutting down. Secure logout complete.
PS C:\Users\KEVIN\OneDrive\Documents\Programing\Java\intern>
```